

CS173: Discrete Structures

Problem Set 4

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Throughout this problem set, we will use the following shorthand notation for axioms:

- **Extensionality** refers to The Axiom of Extensionality (Axiom 1)
- **Pairing** refers to The Axiom of Pairing (Axiom 2)

1. For any set x , the power set of x is given by $\mathbb{P}(x) := \{y \mid y \subseteq x\}$.

(a) Prove $\forall x \forall y (\mathbb{P}(x) \cup \mathbb{P}(y) \subseteq \mathbb{P}(x \cup y))$.

Proof. Let x and y be arbitrary sets.

To prove $\mathbb{P}(x) \cup \mathbb{P}(y) \subseteq \mathbb{P}(x \cup y)$, we must show $\forall w (w \in \mathbb{P}(x) \cup \mathbb{P}(y) \Rightarrow w \in \mathbb{P}(x \cup y))$ by the definition of subset.

Let w be arbitrary and assume $w \in \mathbb{P}(x) \cup \mathbb{P}(y)$.

By the definition of union, $w \in \mathbb{P}(x)$ or $w \in \mathbb{P}(y)$.

Case 1: Suppose $w \in \mathbb{P}(x)$.

By the definition of power set, $w \subseteq x$.

By Problem 1c of PS3, $x \subseteq x \cup y$.

By Problem 1a of PS3 (transitivity of $\subseteq$), $w \subseteq x \cup y$.

Thus $w \in \mathbb{P}(x \cup y)$ by the definition of power set.

Case 2: Suppose $w \in \mathbb{P}(y)$.

By the definition of power set, $w \subseteq y$.

Since $x \cup y = y \cup x$ by commutativity of union, and $y \subseteq y \cup x = x \cup y$ by Problem 1c of PS3, we have $y \subseteq x \cup y$.

By Problem 1a of PS3 (transitivity of $\subseteq$), $w \subseteq x \cup y$.

Thus $w \in \mathbb{P}(x \cup y)$ by the definition of power set.

In both cases, $w \in \mathbb{P}(x \cup y)$.

Therefore $\mathbb{P}(x) \cup \mathbb{P}(y) \subseteq \mathbb{P}(x \cup y)$. □

(b) Prove $\exists x \exists y (\mathbb{P}(x) \cup \mathbb{P}(y) \neq \mathbb{P}(x \cup y))$.

Proof. We exhibit a counterexample. Let $x = \{\emptyset\}$ and $y = \{\{\emptyset\}\}$.

Then $x \cup y = \{\emptyset, \{\emptyset\}\}$, and

$$\mathbb{P}(x \cup y) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}.$$

Meanwhile $\mathbb{P}(x) = \{\emptyset, \{\emptyset\}\}$ and $\mathbb{P}(y) = \{\emptyset, \{\{\emptyset\}\}\}$, so

$$\mathbb{P}(x) \cup \mathbb{P}(y) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}\}.$$

Observe that $\{\emptyset, \{\emptyset\}\} \in \mathbb{P}(x \cup y)$ but $\{\emptyset, \{\emptyset\}\} \notin \mathbb{P}(x) \cup \mathbb{P}(y)$.

Therefore, by Extensionality, $\mathbb{P}(x) \cup \mathbb{P}(y) \neq \mathbb{P}(x \cup y)$ for these choices of x and y , which proves $\exists x \exists y (\mathbb{P}(x) \cup \mathbb{P}(y) \neq \mathbb{P}(x \cup y))$. □

(c) Prove $\forall x \forall y (x \cap y = y \Rightarrow y \in \mathbb{P}(x))$.

Proof. Let x and y be arbitrary sets. Assume $x \cap y = y$.

To show $y \in \mathbb{P}(x)$, we must show $y \subseteq x$ by the definition of power set.

To show $y \subseteq x$, we must show $\forall w (w \in y \Rightarrow w \in x)$ by the definition of subset.

Let w be arbitrary and assume $w \in y$.

Since $x \cap y = y$, we have $w \in x \cap y$.

By the definition of intersection, $w \in x \wedge w \in y$.

Therefore $w \in x$.

Therefore $y \subseteq x$, which means $y \in \mathbb{P}(x)$ by the definition of power set. $\square$

(d) Prove $\forall x \forall y (y \in \mathbb{P}(x) \Rightarrow x \cap y = y)$.

Proof. Let x and y be arbitrary sets. Assume $y \in \mathbb{P}(x)$.

By the definition of power set, $y \subseteq x$, i.e. $\forall w (w \in y \Rightarrow w \in x)$.

To show $x \cap y = y$, we use Extensionality: it suffices to show $\forall w (w \in x \cap y \Leftrightarrow w \in y)$.

Let w be arbitrary.

($\Rightarrow$) Assume $w \in x \cap y$. By the definition of intersection, $w \in y$.

($\Leftarrow$) Assume $w \in y$. Since $y \subseteq x$, we have $w \in x$. Thus $w \in x$ and $w \in y$, so $w \in x \cap y$ by the definition of intersection.

Therefore, by Extensionality, $x \cap y = y$. $\square$

2. For any sets x and y , the **relative complement** of y in x is $x \setminus y := \{z \mid z \in x \wedge z \notin y\}$. The **symmetric difference** of x and y is $x \Delta y := (x \setminus y) \cup (y \setminus x)$.

- (a) Prove that $\forall x ((x \cup \{x\}) \setminus \{x\} = x)$.

Proof. Let x be an arbitrary set.

By Extensionality, it suffices to show $\forall w (w \in (x \cup \{x\}) \setminus \{x\} \Leftrightarrow w \in x)$.

Let w be arbitrary.

($\Rightarrow$) Assume $w \in (x \cup \{x\}) \setminus \{x\}$.

By the definition of relative complement, $w \in x \cup \{x\}$ and $w \notin \{x\}$.

By the definition of union, $w \in x$ or $w \in \{x\}$.

Since $w \notin \{x\}$, we must have $w \in x$.

($\Leftarrow$) Assume $w \in x$.

By Problem 1c of PS3, $x \subseteq x \cup \{x\}$, so $w \in x \cup \{x\}$.

Since $x \notin x$ by Clavicula Theorem 5.1, and $w \in x$, we have $w \neq x$, so $w \notin \{x\}$.

Therefore $w \notin \{x\}$, and so $w \in (x \cup \{x\}) \setminus \{x\}$.

Therefore, by Extensionality, $(x \cup \{x\}) \setminus \{x\} = x$. □

- (b) Prove that $x \Delta y = (x \cup y) \setminus (x \cap y)$ for all sets x and y .

Proof. Let x and y be arbitrary sets.

By Extensionality, it suffices to show $\forall w (w \in x \Delta y \Leftrightarrow w \in (x \cup y) \setminus (x \cap y))$.

Let w be arbitrary.

Note that by the respective definitions:

$$\begin{aligned} w \in x \Delta y &\Leftrightarrow w \in (x \setminus y) \cup (y \setminus x) \\ &\Leftrightarrow (w \in x \wedge w \notin y) \vee (w \in y \wedge w \notin x). \end{aligned}$$

$$\begin{aligned} w \in (x \cup y) \setminus (x \cap y) &\Leftrightarrow w \in x \cup y \wedge w \notin x \cap y \\ &\Leftrightarrow (w \in x \vee w \in y) \wedge \neg(w \in x \wedge w \in y) \\ &\Leftrightarrow (w \in x \vee w \in y) \wedge (w \notin x \vee w \notin y). \\ &\Leftrightarrow (w \in x \wedge w \notin y) \vee (w \in y \wedge w \notin x). \end{aligned}$$

Since both conditions reduce to $(w \in x \wedge w \notin y) \vee (w \in y \wedge w \notin x)$, we have $w \in x \Delta y \Leftrightarrow w \in (x \cup y) \setminus (x \cap y)$.

Therefore, by Extensionality, $x \Delta y = (x \cup y) \setminus (x \cap y)$. □

3. Recall that the Axiom of Regularity asserts $\forall x (x \neq \emptyset \Rightarrow \exists y (y \in x \wedge x \cap y = \emptyset))$.

(a) Show that $\forall x (x \neq x \cup \{x\})$.

Proof. Let x be an arbitrary set. Consider the set $\{x\}$, which exists by Pairing. Suppose for contradiction that $x = x \cup \{x\}$.

Since $\{x\} \neq \emptyset$, by the Axiom of Regularity we have some $y \in \{x\}$ such that $\{x\} \cap y = \emptyset$.

Since $y \in \{x\}$, we have $y = x$.

Thus $\{x\} \cap x = \emptyset$.

Recall $x \in x \cup \{x\}$, since $x = x \cup \{x\}$ then $x \in x$.

Since $x \in \{x\}$ and $x \in x$, we have $x \in \{x\} \cap x$.

This contradicts $\{x\} \cap x = \emptyset$.

Therefore, by contradiction, $x \neq x \cup \{x\}$. □

(b) Show that $\forall x \forall y (x \neq y \Rightarrow x \cup \{x\} \neq y \cup \{y\})$.

Proof. Let x and y be arbitrary sets such that $x \neq y$.

Suppose for contradiction that $x \cup \{x\} = y \cup \{y\}$.

Since $x \in \{x\} \subseteq x \cup \{x\} = y \cup \{y\}$, we have $x \in y \cup \{y\}$.

By the definition of union, $x \in y$ or $x \in \{y\}$. Since $x \in \{y\}$ implies $x = y$, contradicting $x \neq y$, we must have $x \in y$.

By the same argument with x and y swapped, $y \in x \cup \{x\} = y \cup \{y\}$ gives $y \in x$.

Now consider the set $\{x, y\}$, which exists by Pairing. It is nonempty, so by the Axiom of Regularity there exists $z \in \{x, y\}$ such that $\{x, y\} \cap z = \emptyset$.

Since $z \in \{x, y\}$, either $z = x$ or $z = y$.

If $z = x$, then $\{x, y\} \cap x = \emptyset$. But $y \in x$ and $y \in \{x, y\}$, so $y \in \{x, y\} \cap x$, a contradiction.

If $z = y$, then $\{x, y\} \cap y = \emptyset$. But $x \in y$ and $x \in \{x, y\}$, so $x \in \{x, y\} \cap y$, a contradiction.

In both cases we reach a contradiction, so $x \cup \{x\} \neq y \cup \{y\}$. □

4. We define the **Cartesian product** of two sets X and Y by $X \times Y := \{(x, y) \mid x \in X \wedge y \in Y\}$. Show that the Cartesian product of any two sets exists.

Proof. Let X and Y be arbitrary sets.

We first show that every ordered pair (x, y) with $x \in X$ and $y \in Y$ is an element of $\mathbb{P}(\mathbb{P}(X \cup Y))$.

Let $x \in X$ and $y \in Y$. Since $x \in X$ and $y \in Y$, both x and y are in $X \cup Y$. Therefore $\{x\} \subseteq X \cup Y$ and $\{x, y\} \subseteq X \cup Y$, so $\{x\}, \{x, y\} \in \mathbb{P}(X \cup Y)$. Thus $(x, y) = \{\{x\}, \{x, y\}\} \subseteq \mathbb{P}(X \cup Y)$, which means $(x, y) \in \mathbb{P}(\mathbb{P}(X \cup Y))$.

By the Axiom of Power Set, $\mathbb{P}(X \cup Y)$ and $\mathbb{P}(\mathbb{P}(X \cup Y))$ both exist.

Let $\varphi(z) := \exists x \exists y ((z = (x, y)) \wedge x \in X \wedge y \in Y)$. By the Axiom of Separation, we obtain the set

$$X \times Y := \{z \in \mathbb{P}(\mathbb{P}(X \cup Y)) \mid \exists x \exists y ((z = (x, y)) \wedge x \in X \wedge y \in Y)\}.$$

This set contains exactly the ordered pairs (x, y) with $x \in X$ and $y \in Y$, so $X \times Y$ exists. $\square$